



Carlos Carranza López

Software developer | Cybersecurity engineer

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Software developer with a Master's specialisation in cybersecurity, focused on building secure, well-engineered systems end-to-end – from full-stack web applications to provenance and authentication tooling. Comfortable moving between development and security research, with hands-on experience in SDN, PKI/authentication, malware analysis, and AI-content provenance (C2PA/JPEG Trust).

Education

Double Bachelor Degree Engineering — Computer Engineering & Telecom Systems

2020 – 2025

Universitat Autònoma de Barcelona (UAB)

Key Subjects: *Security:* Information & Security, Information Assurance & Security | *Networks:* Telecom Networks, Advanced Internet Technologies | *Software:* Software Engineering & Design, Distributed Systems, Web Technologies | *AI & Signals:* Artificial Intelligence, Digital Signal Processing

Final Degree Projects (TFG):

- Full-stack web development for *Viver de Belloc* [TFG]
- Antenna data acquisition in anechoic chamber (spectrum analyser & motor control via SCPI) [TFG]

M.Sc. — Cybersecurity

2025 – 2026

Universitat Politècnica de Catalunya (UPC)

Master's Thesis (TFM):

- Provenance Information for AI-Generated Image

Relevant Coursework:

- Software Defined Security (SDN, NFV, network telemetry)
- Network Security — Authentication & Authorisation (PKI, 2FA, tokens)
- Cybersecurity Use Cases (Reconnaissance, OSINT, Exploitation, Persistence)

Work Experience

Web Developer — Gamification Platform

2025

Dept. d'Enginyeria de la Informació, UAB

- Designed and built, end-to-end, a full-stack challenge-based gamification platform used by the department to track student progress.

Full-Stack Developer — Viver de Belloc

2024 – 2025

Freelance

- Built and shipped the company's website from scratch as sole developer, working directly with the client from requirements to deployment.

Arduino Workshop Instructor

2018

Primary & Secondary Schools

- Designed and delivered hands-on programming and electronics workshops for students aged 4–6, introducing early computational thinking through Arduino.

Projects

JPEG Trust Orchestrator

2026

Java Spring service that generates, signs, and chains C2PA/JPEG Trust provenance manifests for AI-generated images, normalising heterogeneous sources (DALL-E, Firefly, Stable Diffusion, Gemini) into a single interoperable schema. [GitHub] *Java, Spring, C2PA*

C2PA Reader

2026

Python/Flask server that extracts and validates C2PA manifests from images, exposing results through REST endpoints for downstream tools. *Python, Flask, C2PA*

Real-Time Browser Inspector

2026

Browser extension that detects AI images on any webpage and displays their C2PA/JPEG Trust provenance status on hover,

with colour-coded trust indicators. *JavaScript, WebExtensions API*

Malware Analysis Project

2025

Security research on malware behaviour and detection techniques. [GitHub] *Python, YARA*

Spectrum Analyser & Motor Controller

2025

Python program using SCPI commands to control a spectrum analyser and motor, collecting and processing antenna-array data in an anechoic chamber (TFG). *Python, SCPI*

NASA Space Apps Challenge 2025 — Leonidas

2025

International hackathon project built under time pressure with a cross-functional team. [GitHub] *Vue.js*

Technical Skills

Languages: Python, C++, JavaScript, Java, SQL, Dart, Bash

Web: Vue.js, HTML, CSS, REST APIs, Flutter

Cloud: AWS, Cloud Computing

Security Tools: Wireshark, ClamAV, YARA, CVSSv4, John the Ripper, Mininet

Cybersecurity: OSINT, Recon, Exploitation, Persistence, SDN Security, Zero Trust, PKI, 2FA

Dev Tools: VS Code, IntelliJ, Git, LaTeX, MATLAB

Other: SCPI, Arduino, IBM Quantum, AR/3D

Languages

Spanish Native (C2)

Catalan Native (C2)

English Upper-Intermediate (B2)

Hackathons

◇ UPC Hackathon 2023

◇ NASA Space Apps Challenge 2025

Interests

★ Manga & novels

★ Cinema

★ CTF & security challenges

★ Tech industry news